

Wi-Fi Security Protocol

Module 32

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Video: <https://www.youtube.com/watch?v=vCdk5ufJpPo>

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana today in this lecture we will study about WiFi security protocols first of all please see the contents which I shall cover after covering brief give introduction about three basic Wi-Fi security protocols I shall cover each one by one first in this series will be wired equivalent privacy we followed by WiFi protected access WPA and WPA2 in the last I shall compare WPA and WP pa2 technology the wireless networking is becoming very popular among large number of Internet users because of the popularity of the wireless networking technology large number of wireless networking products and Protocols are

available for the home and Business use wireless Transmissions are available to authorized users and also to the unauthorized users like hackers the i e 802.11 standard offers some level of protection this protection known as the wired equivalent privacy W protocol defines a set of instructions and rules by which wireless data can be transmitted over the air waves with some amount of security wired equivalent privacy W protocol is a basic security feature in the i e 802.11 standard intended to provide confidentiality over a wireless network by encrypting information sent over the 3 minutes, 1 second network very soon web got replaced by Wi-Fi protocol access WPA due to a flaw in web of getting cracked in a few minutes with the aid of

automated tools WiFi protected access WPA and WPA2 is a wireless security protocol designed to address and fix the known security issues in we anyway this was very brief about basic security techniques used for wifi now let us see each one by one in detail first is wired equivalent privacy web which is a security algorithm for i e a 802.11 wireless networks introduced as part of the original 802.11 standard ratified in 1997 its intention was to provide data confidentiality comparable to that of a traditional wired Network the design objectives of web are as per i e 82. 11 standard which states the following reasonably strong the security afforded by the algorithm should rely on the difficulty of discovering the secret key

through a Brute Force attack this in turn is related to the length of the secret key and the frequency of changing Keys self synchronizing web should be self- 5 minutes, 1 second synchronizing for each message this property is critical for a data link level encryption algorithm where best effort delivery is assumed efficient the web algorithm should be efficient and may be implemented in either Hardware or software exportable every effort should be made to design the web system operation so as to maximize the chances of approval by the US government for exporting this technology optional the implementation and use of web in I 802.11 should be optional from the above objectives it's clear that VP was not designed to provide a high military level

security the intention was to make it hard to break in as opposed to impossible to break in now let us move on to the technical details of web web security involves two Parts authentication and encryption authentication in web involves authenticating a device when it first joins the Lan the authentication process in the wireless network using web is to prevent devices or stations joining the network unless they know the web 7 minutes, 1 second key two methods of authentication can be used with web Open System Authentication and shared key authentication in Open System Authentication the W Lan client need not provide its credentials to the access point during authentication any client can authenticate with the access point and then attempt to associate in fact no

authentication occurs subsequently VB keys can be used for encrypting data frames at this point the client must have the correct keys in shared key authentication the web key is used for authentication in a for Four Step Challenge response handshake as shown in figure step one wireless device sends authentication request to the wireless access point step two in response wireless access point sends 128bit random authentication Challenge in a clear text to the requesting Wireless device step three the wireless device uses the shared secret key to sign the challenge and send them in form of authentication response to the wireless access point step four wireless access point decrypts the signed message using the

shared secret key and verifies the challenge that that it has sent 9 minutes, 1 second before if the challenge matches then authentication succeeds otherwise not if authentication succeeds then authentication success message is sent from access point to wireless device in webbased security unfortunately no secret key is exchanged after Authentication the same secret key or shared key is used for both authentication and encryption so there is no way to tell whether the subsequent message comes from the trusted device or from an imposter this kind of authentication is prone to man in the middle attack next we shall take up encryption part of web please see figure which shows stepbystep implementation of encryption process web uses rc4 stream Cipher to

encrypt data between access point and wireless device web uses 8bit rc4 and operates on 8bit values by creating an array with 256 8 bit values for a lookup table VB uses CRC for the data integrity and accordingly performs cyclic redundancy check operation on the plain text and

generates CRC value the CRC value is concatenated to the plain text the secret key is concatenated to the initialization Vector IV and fed into the RC4 based on the secret key and IV RC4 generates key stream the key stream and plain Text Plus CRC message are XORed together the result is the cipher text the same initialization Vector that was used before is prepended in clear text to the resultant Cipher text the initialization Vector IV plus Cipher text along with the frame headers

are then transmitted over the air in this encryption method as we have seen some part of the secret key is used with different exposed values and attacker can generate the secret part by analyzing some portion of number of bits in the first few bytes of the key stream with relatively less work another problem with WEP is that the secret shared key is concatenated with the visible IV value this weakness is known as IV weakness in the beginning it was believed that IV offers impenetrable resistance to the eavesdroppers and hackers however as wireless networks began to grow in popularity many Cryptanalysts and researchers discovered flaws in the original WEP design many believe that there was little peer review performed on the WEP

protocol many of the WEP flaws would have been caught in the early design phase if its design and implementation specifications had been reviewed thoroughly for most of the wireless networking users especially home users WEP is the only choice available until new security mechanisms are added to the IEEE 802.11 standard but as people say something is better than nothing even with its known weakness WEP was still more effective than no security at all at least it will provide some security against unauthorized use of wireless LANs, 1 second network and eating up the bandwidth there are many solutions available to overcome the weakness of WEP some of them are the bigger size of the initialization Vector IV can be chosen the hashed value of IV can be

prepended or appended to the cipher text instead of the clear text instead of using CRC checksum different method can be used for the data Integrity verification that is Hash functions change secret key regularly dynamically using secure symmetric key distribution protocols better key management using security handshake protocols new authentication mechanisms using the extensible Authentication Protocol EAP in spite of implementation of these improvements there was a growing trend of getting cracked in a few minutes with the aid of automated tools so instead of just relying on the WEP security alone additional measures must be taken to provide better security among wireless devices in 2003 the Wi-Fi Alliance

announced that WEP had been superseded by Wi-Fi protected access WPA in 2004 with a ratification of the full IEEE 802.11 standard that is WPA2 the IEEE declared that WEP has been declared obsolete Wi-Fi Alliance also decided to implement IEEE 802.11i standard to overcome the problems in WEP IEEE 802.11i was implemented in two steps Wi-Fi protected access WPA and Wi-Fi protected access 2 that is WPA2 the Wi-Fi Alliance intended WPA as an intermediate measure to take the place of WEP pending the availability of full IEEE 802.11i standard WPA

could be implemented through firmware upgrades on wireless network interface cards designed for web as WPA was an intermediate solution for Hardware that could not support WPA 2 so

to implement WPA an interim software implementable solution was designed that could delay immediate deployment of new hardware to add support for WPA or WPA 2 some old Wi-Fi access points might need to be replaced or have their firmware upgraded WPA which is sometime referred to as draft 11 standard became available in 2003 WPA2 became available in 2004 and is commonly known as full 11i or 11i 2004 standard WPA is a wireless security protocol designed to address and fix the known security issues in web WPA provides users with a higher level of assurance that their data will remain protected by using temporal key Integrity protocol (TKIP) for data encryption specifically the temporal key

Integrity protocol (TKIP) was adopted for WPA we used a 40 bit or 104 bit encryption key that must be manually entered on wireless access points and devices which does not change but TKIP employs a per packet key meaning that it dynamically generates a new 128bit key for each packet and thus prevents the type of attack that compromise WEP WPA also includes a message Integrity check which is designed to prevent an attacker from altering and resending data packets this replaces the cyclic redundancy check (CRC) that was used by the WEP standard CRC's main flaw in WEP was that it did not provide a sufficiently strong data Integrity guarantee for the packets it handled well tested message authentication codes existed to solve

these problems but they required too much computation to be used on Old network cables WPA uses a message Integrity check algorithm called Michael to verify the Integrity of the packets no doubt Michael is much stronger than a CRC but still researchers have discovered a flaw in WPA due to reason of limitations of Michael to retrieve the key stream from short packets to use for reinjection and spoofing in order to solve such problems 21 minutes, 1 second in WPA and to completely implement 11i protocol a new wireless security protocol WPA2 is implemented in which only authorized users can access a wireless device with features supporting stronger cryptography stronger Authentication control key management replay attack protection and data

Integrity going into technical aspects of WPA 2 instead of TKIP advanced encryption standard AES is used for encryption purpose that has a fixed block size of 128 bits and three different key sizes using three different steps of the algorithm in the first iteration a 128bit key is used to perform 9 rounds in the second a 192bit key performs 11 rounds and in the third iteration a 256bit key is used to perform 13 rounds because AES is a substitution Cipher So within each round bits are substituted and rearranged and then special multiplication is performed based on the new Arrangement the effectiveness of AES cannot be disputed as the time needed to break it by using a Brute Force attack with a 128bit key length is around 2200

years now after brief technical details let us compare WPA and WPA 2 there are some similarities and differences between WPA and WPA 2 both the security techniques WPA and WPA 2 uses the 802.1x that is EAP framework as part of the infrastructure that ensures Central Mutual authentication and dynamic Key Management both offers a pre-shared key for using home and small office environments and both are designed to secure all versions of 802.11 devices including 802.11b 802.11 A and 802.11 G now coming onto the differences in both techniques WPA2 uses a mix mode that supports both WPA and WPA2 enabled devices on the same wireless network but WPA supports only one however the most significant difference between WPA and WPA2 is that

WPA2 uses the advanced encryption standard AES instead of tki I for data 25 minutes, 1 second encryption WPA2 is theoretically not hackable but WPA is WPA2 requires more processing power than WP a so dear friends today in this lecture we have covered introductory concepts of wi-fi security followed by by three basic security techniques VB WPA and WPA 2 used for Wi-Fi in last we have also seen comparison of two latest security techniques WPA and WPA 2 so this was all about WiFi security protocols thank you e